



LVT Manufacturer in South Korea **KCC GLASS**



COMPANY PROFILE

Under the management principle of “Creating value for a better life”, KCC Glass Corporation hopes to pursue management focused on value, creation, trust and environmental friendliness that satisfies our customers, shareholders, the society as well as our company members. Based on your trust, we will continue to influence and lead the global market in addition to achieving our company’s vision through bold and innovative R&D.

CAPACITY
(Per year)

6,000,000M²

NUMBER OF EMPLOYEES
(Unit : persons)

2,045

SALES
(USD)

1.3B

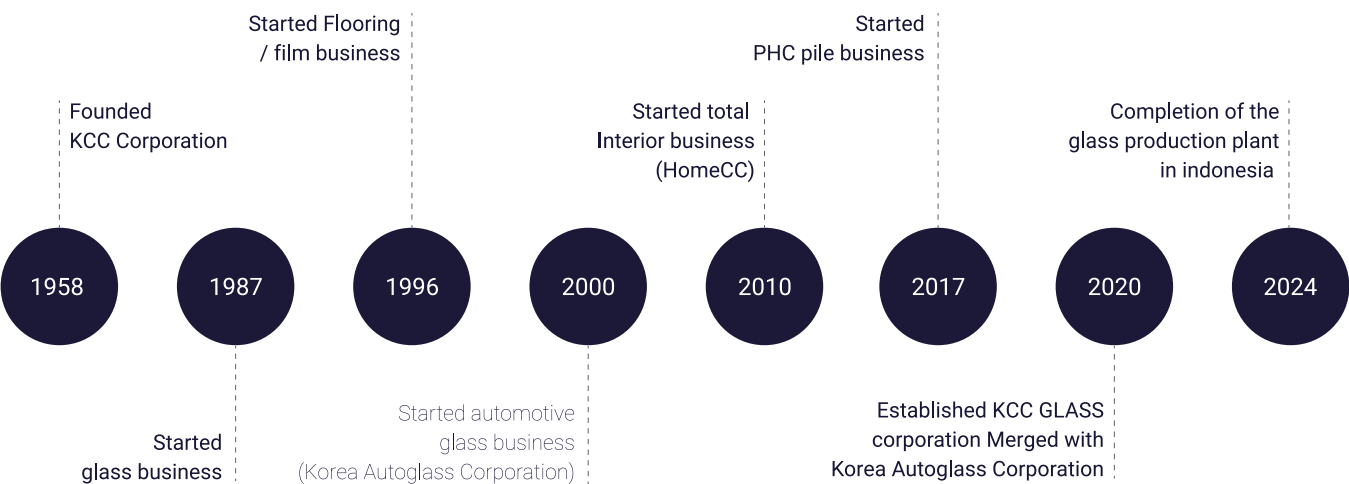
COMPANY OVERVIEW

KCC GLASS Corporation was established in January 2020 with the expertise in glass and interior products include flooring, film and interior stone as a spin-off from KCC Corporation which is one of the oldest and largest construction material manufacturing company in Korea.

VISION

Number one company that cherishes customer value and leads the global market with green technologies

KEY MILESTONES



PRODUCTION CAPABILITY(for OEM)

PLANK

Type	Type of Flexible LVT	Thickness (mm)	Wear Layer (mm)	Glass Fiber (GF)	Remark	Available Size List		
						Inch	mm	
PLANK	Dryback (Glue Down)	2	0.15	N/A		7.24" X 37.4"	184 X 950	
			0.2			7.25" X 48"	184.15 X 1219.2	
			0.3			6" X 36"	152.4 X 914.4	
		2.5	0.15			6" X 48"	152.4 X 1219.2	
			0.2			7" X 48"	177.8 X 1219.2	
			0.3			9" X 48"	228.6 X 1219.2	
			0.5			9" X 60"	228.6 X 1524	
			Total 7 Options					
			3		0.15			
					0.2			
		0.3						
		0.5						
		4.5	0.55					
	0.5							
	Looselay (Option Anti-Slip or Acoustic)	5	0.5	Included	SENSELAY			
			0.55					
			0.7					

TILE

Type	Type of Flexible LVT	Thickness (mm)	Wear Layer (mm)	Glass Fiber (GF)	Remark	Available Size List	
						Inch	mm
PLANK	Dryback (Glue Down)	2	0.15	N/A		18" X 18"	457.2 X 457.2
			0.2			18" X 36"	457.2 X 914.4
			0.3			23.6" X 23.6"	600 X 600
		2.5	0.15			Total 3 Options	
			0.2				
			0.3				
			0.5				
			0.15				
		3	0.2				
			0.3				
			0.5				
			0.55				
	Looselay (Option Anti-Slip or Acoustic)	4.5	0.5	Included	SENSELAY		
			0.5				
		5	0.55				
			0.7				

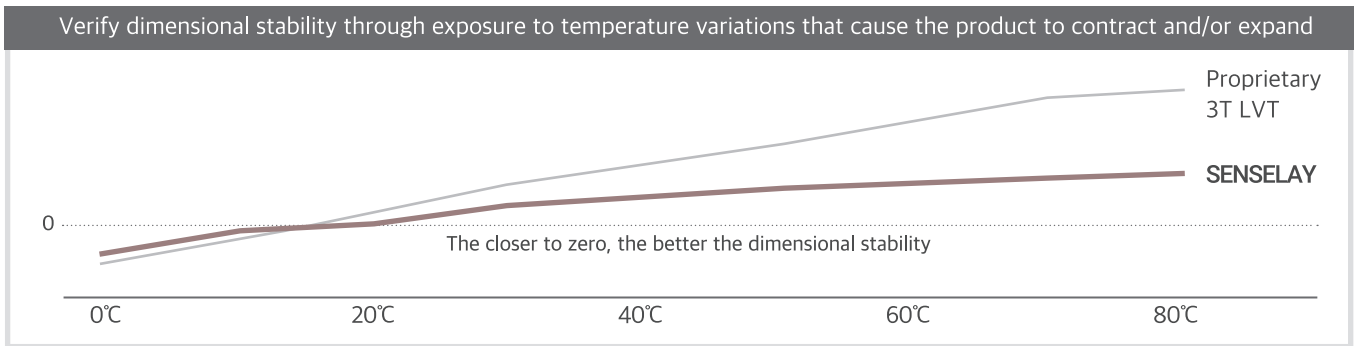
* Different thicknesses of wear layer are available with the order base.

ANNEALING SYSTEM

KCC Glass’s Egis Annealing System, which set a record for being the longest in Korea, helps minimize any deformities in the flooring structure from temperature/humidity fluctuation while maximizing dimensional stability. The Egis Annealing System ensures that materials cool slowly over a set period of time and that any change from temperature variations are finalized to achieve a stable equilibrium in floorings. This results in high-quality flooring less susceptible to dimensional fluctuations or other defects under varying conditions. The glass fiber layer produced by applying PVC SOL coating to glass fiber renders our products even more resistant to deformities to further upgrade dimensional stability.

※ Glass Fiber Glass fiber is made by melting glass and pulling it into thin, long strands

Dimensional Stability Test



* Test method: Raise the temperature by 10°C every hour within the range of 0°C~80°C, and record any change

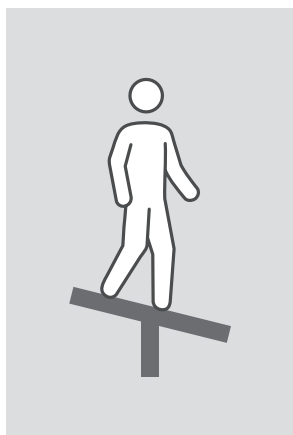
NON-SLIP

Non-slip surface reduces the risk of slips and/or falls. This design process focuses on structural safety to help people at locations such as child education institutions, multi-use facilities and healthcare facilities to avoid a wide array of hazardous conditions while going about their everyday activities.

SLIP TEST

- The subject wears a pair of lubricated test shoes and stands vertically to measure the angle of slip.

RAMP TEST RESULTS(DIN 51130:2014): **R11** rating (highest)

	Test on an "inclined plane"		Commercial area
	Assessment Groups	Angle of inclination	Graphic
	R9	> 3° - 10°	
	R10	> 10° - 19°	
	R11	> 19° - 27°	
	R12	> 27° - 35°	
	R13	> 35°	

* Results may vary slightly depending on the type of product embossing that is being tested, along other factors.

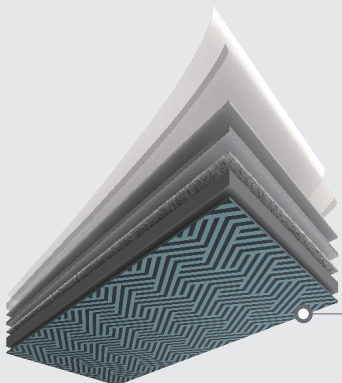
NOISE MITIGATION

Your tranquil space insulated through our Acoustic Layer technology

Acoustic Layer, an eco-friendly cushioning layer that comes complete with KCC Glass's unique high-performance Egis Comfort Technology, reduces lightweight sound impact of the floor to reduce living noise and add a layer of sound protection to your environment.

The Acoustic Layer absorbs large impacts while smoothing out undulating floor surfaces to provide enhanced step comfort. Your tranquil space is insulated through our Acoustic Layer technology.

Egis Comfort Tech



Balance Layer

***Acoustic Layer**

Apply the Acoustic Layer which functions as a cushioning layer at the very bottom of the product to reduce impact noise

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Non-Acoustic

Acoustic



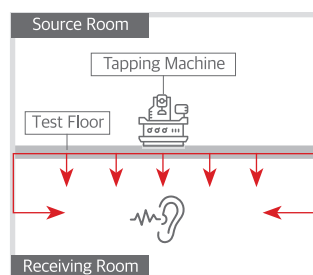


- Apply eco-friendly PVC form at the very bottom of the product
- Smooth out undulating floor surfaces with flexible cushioning materials
- Gently absorb the shock applied to the flooring to reduce everyday noise and deliver exceptional comfort

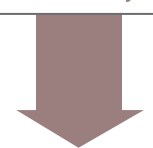
- Apply LooseLay pattern, the unique feature of KCC Glass's Acoustic Layer, to provide excellent flooring settlement

※ The Acoustic Layer is optional, and is PVC-integrated.

SOUND INSULATION TEST (ISO 717-2)



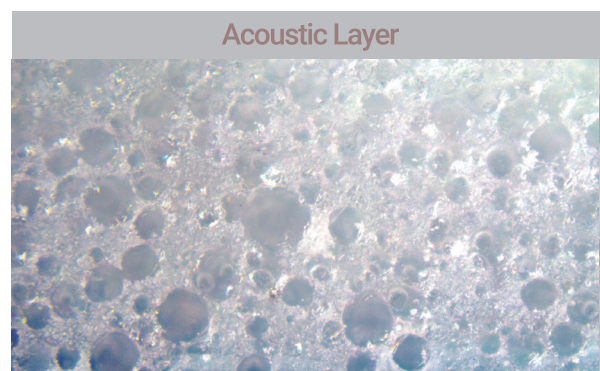
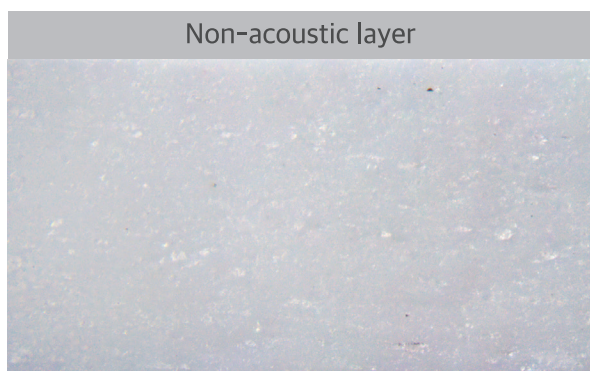
SENSELAY 5.0T
Acoustic Layer



13dB

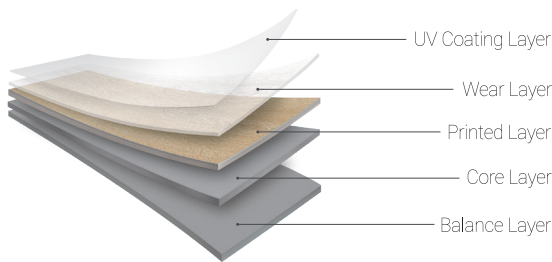
- Install products on a floor with a standard-based building and measure the level of sound insulation on the lower floor when noise is generated

※ The Acoustic Layer is optional, and is PVC-integrated.

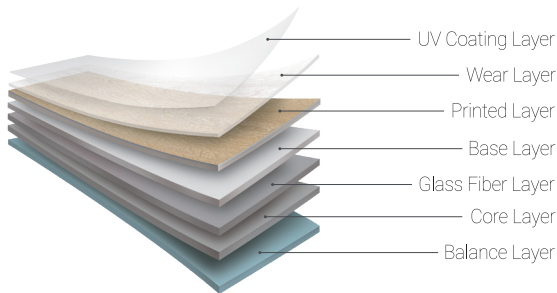


TEST REPORT _ ASTM

Glue down



Looselay

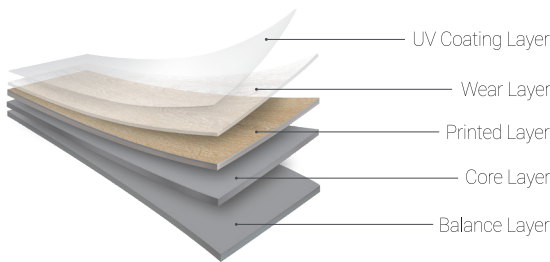


• ASTM

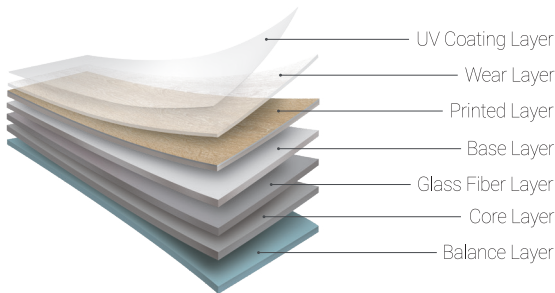
Test Item	Standard	Results	Test Method
Thickness	As specified ± 0.005 in./lin.ft(0.13mm)	PASS	ASTM F386
Wear Layer Thickness	-	PASS	ASTM F410
Size and Squareness : Size	± 0.016 in./lin.ft(0.4mm/305mm)	PASS	ASTM F2055
Size and Squareness : Squareness	≤ 0.010 in(0.25mm)	PASS	ASTM F2055
Flexibility	1 inch mandrel - no crack or break	PASS	ASTM F137
Dimensional Stability and curl	≤ 0.020 in./lin.ft(0.51mm/305mm)	PASS	ASTM F2199
Residual Indentation	Average < 8%	PASS	ASTM F1914
Static Load Limit	≤ 0.005 "(0.127mm)@250psi	PASS	ASTM F970
Resistance to Heat	Average & Max ΔE < 8.0	PASS	ASTM F1514
Resistance to Light	Average & Max ΔE < 8.0	PASS	ASTM F1515
Chemical Resistance	No more than Slight Change	PASS	ASTM F925
Slip Resistance	ADA Compliant(> 0.50)	PASS	ASTM C1028
Slip resistance-james machine	≥ 0.50	PASS	ASTM D2047
Slip resistance-pendulum test	dry/wet ≥ 31	PASS	ASTM E303
Flammability	Cass I (>0.45 W/cm2)	PASS	ASTM E648
Smoke Density (Flaming / Non Flaming)	< 450	PASS	ASTM E662

TEST REPORT _ EN

Glue down



Looselay



• EN

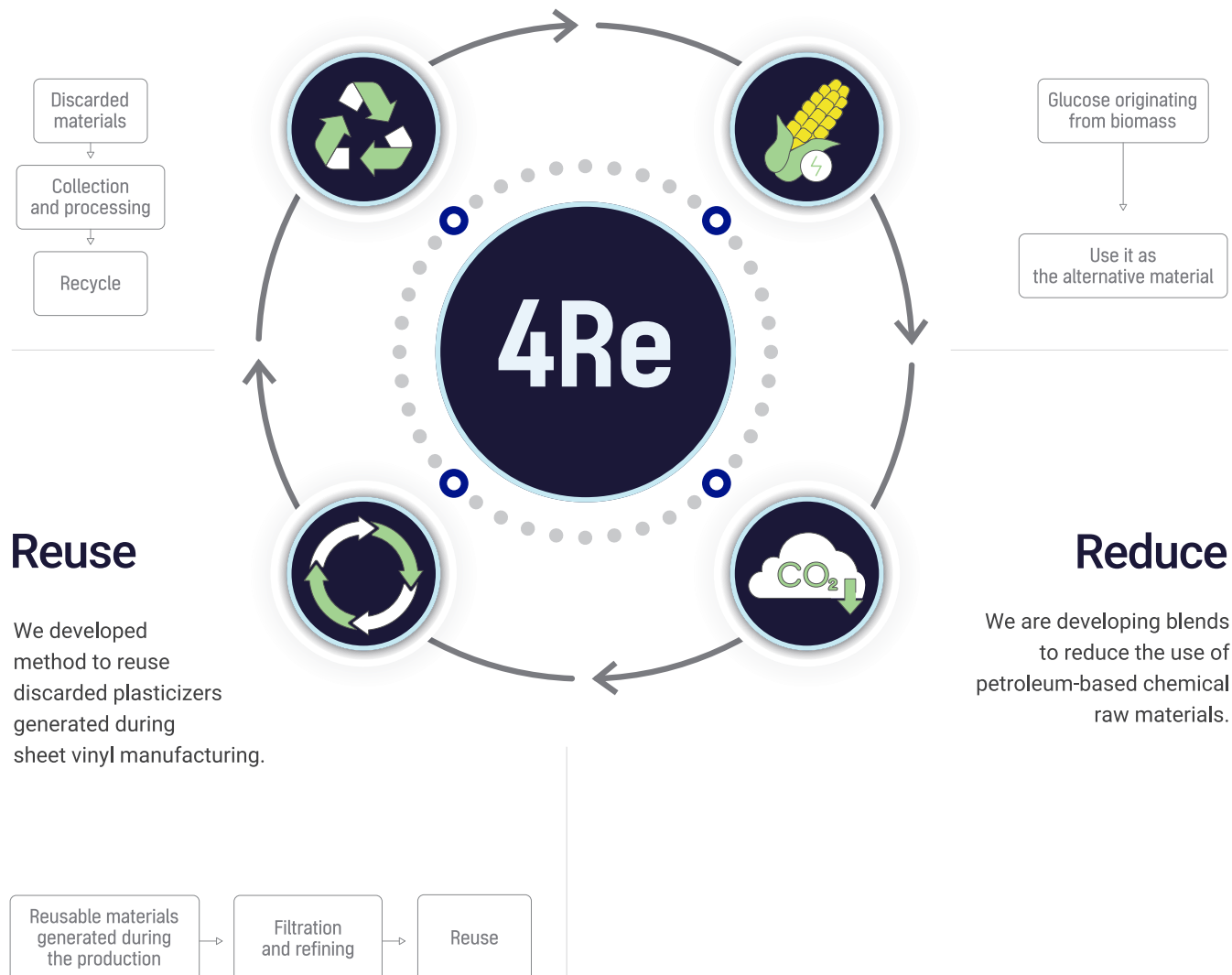
Test Item	Standard	Results	Test Method
Overall Thickness	Nominal value +0.13, -0.10mm	PASS	EN ISO 24346
Wear Layer Thickness	+13%, -10%	PASS	EN ISO 24340
Size and Squareness & Straightness	≤400mm : ≤0.25mm / >400mm : ≤0.35mm	PASS	EN ISO 24342
Wear Resistance	Wear Group : P/T	PASS	EN 660-2
Flexibility	No crack	PASS	EN ISO 24344
Dimensional Stability after exposure to heat	≤0.25%	PASS	EN ISO 23999
Curling after exposure to heat	≤2mm	PASS	EN ISO 23999
Residual Indentation	Average ≤ 0.1mm	PASS	EN ISO 24343-1
Castor Chair Resistance	No Change	PASS	EN ISO 4918
Resistance to Staining	No Change	PASS	EN ISO 26987
Thermal Resistance	0.03~0.04m2·K/W	PASS	EN 12667
Light Fastness	≥6	PASS	EN ISO 105-B02
Slip Resistance	R9 ~ R11	PASS	DIN51130
Dynamic Coefficient of friction	Class DS (>0.30)	PASS	EN 13893
Reach	N.D	PASS	SVHC

KCC GLASS'S ECO FRIENDLY SOLUTIONS

Recycle

We are preparing flooring materials made from plasticizers extracted from discarded PET bottles.

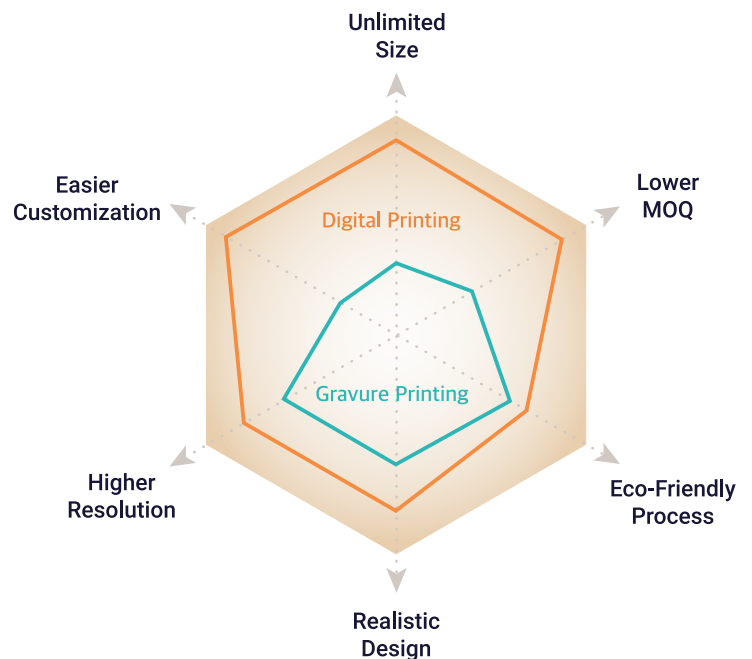
We are currently working on method to recycle discarded wallpaper and ways to utilize it into the Core Layer.



INNOVATION _ DIGITAL PRINTING

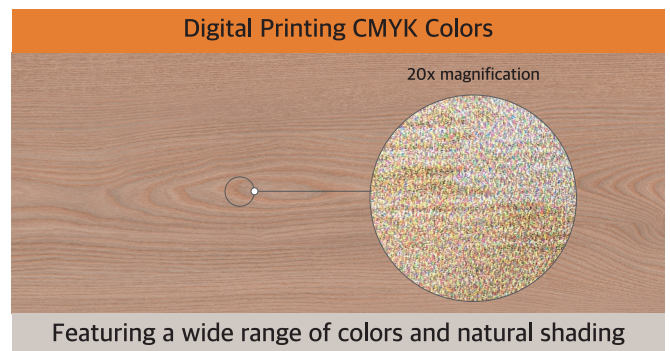
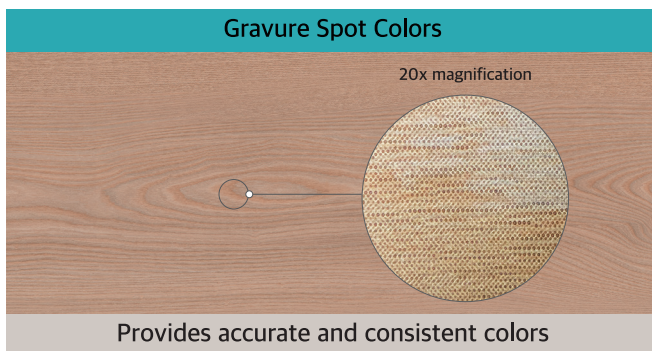
KCC GLASS Corporation is creating new industrial standards for digital UV inkjet printing technologies for interior film and flooring.

We will be a game-changer in the digital printing market based on the best quality and productivity. Our optimized UV inks for interior film and flooring products ensure durability and weather resistance.



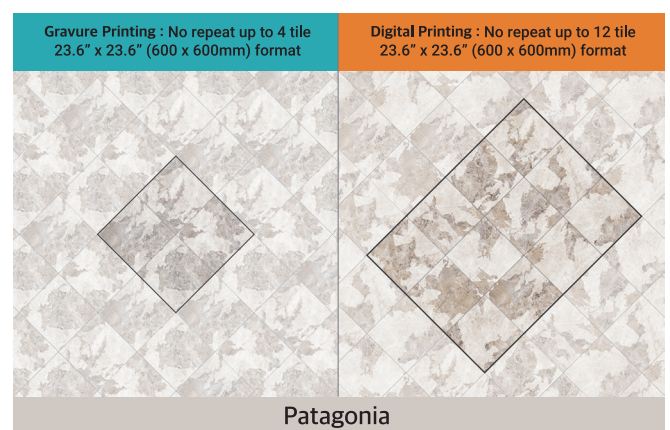
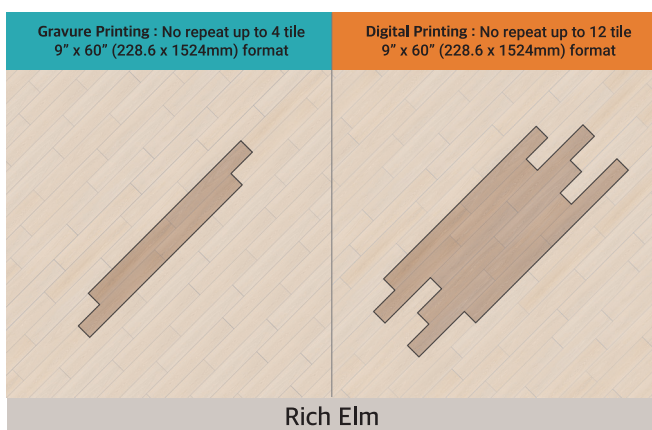
• Enhanced Visual Clarity

KCC GLASS Corporation's digital printing allows for sharper and more delicate expressions compared to traditional gravure printing. By applying the CMYK color model instead of gravure spot colors, it creates a wide color spectrum.



• Minimized Pattern Repetition

Minimized pattern repetition enhances natural and vivid representations, showcasing appearances similar to real wood grain and natural stone.



DESIGN SOLUTION

PREMIUM DESIGN

Creative flooring design, based on extensive knowledge and expertise.

ARTWORKS / PRINT ROLLS

1,300

EMBOSS PATTERNS

250

• Core Competency



Customer-Centered Service

We provide bespoke designs for every project to fulfill our customers' requests. And we constantly strive to design differentiated services to fit your needs.



Numerous Design Assets

Since 1996, we have been running the business of PVC flooring, Laminated film for construction, and VCM for Home appliances. Until today, we continuously collect and design original artworks.



In-House System

We are working as a systematic group based on our 30 years of knowledge and background. And we are a group of designers who can design patterns, work with color separation, and match colors on our own.



Trend Analysis

We consistently forecast and analyze global trends every year. And our latest collections are the results of this research.



ESG

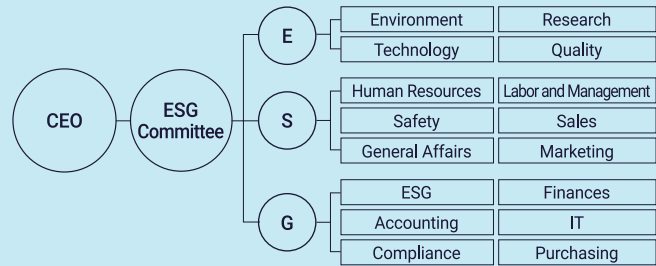
ESG Organization

ESG Organizational Structure

In order to internalize ESG management throughout the entire value chain of the company, KCC GLASS has organized the ESG Committee and Internal Operation Regulations.

Through the ESG Committee, which consists of leaders of each department, we are encouraging company-wide collaboration and active participation in ESG issues.

ESG Committee Organizational Chart



ESG Vision and Strategy



• CERTIFICATION

Product Certification



System Certification



ESG Rating





Corporate Headquarters
587 Gangnam-daero, Seocho-gu, Seoul